THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET C

Chapters 1 & 2: **A Square and a Cube, Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a perfect square? 1

(a) 82 **(b)** 90 **(c)** 120 **(d)** 81

2. The value of 36^2 is: 1

(a) 1260 **(b)** 1332 **(c)** 1296 **(d)** 72

3. $\sqrt{676}$ is equal to: 1

(a) 26 **(b)** 27 **(c)** 25 **(d)** 28

4. The sum of the first 12 odd natural numbers is: 1

(a) 143 **(b)** 24 **(c)** 156 **(d)** 144

5. How many non-square numbers lie between 11^2 and 12^2? 1

(a) 21 **(b)** 23 **(c)** 11 **(d)** 22

6. $7^3 \times 7^3$ equals: 1

(a) 7^6 **(b)** 49^6 **(c)** 7^9 **(d)** 7^0

7. $5^8 \div 5^3$ equals: 1

(a) 5^2 **(b)** 5^{11} **(c)** 5^{24} **(d)** 5^5

8. $(5^2)^3$ equals: 1

(a) 5^5 **(b)** 5^8 **(c)** 10^3 **(d)** 5^6

9. The value of 3^{-2} is: 1

(a) $\frac{1}{6}$ **(b)** 9 **(c)** $\frac{1}{9}$ **(d)** -9

10. The value of 2^{-5} is: 1

(a) $\frac{1}{32}$ **(b)** -32 **(c)** $\frac{1}{10}$ **(d)** 32

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. How many numbers lie between 15^2 and 16^2? 2

12. Find $\sqrt{1521}$ by prime factorisation. 2

13. Simplify $\dfrac{10^5 \times 10^{-2}}{10^2}$. 2

14. Express $\dfrac{1}{64}$ as a power of 2 with a negative exponent. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the least square number divisible by each of 6, 8 and 12. 3

16. A bacterium doubles every hour. Starting with one bacterium, how many are there after 10 hours? Write as a power. 3

17. Is 1620 a perfect square? Use prime factorisation to decide. 3

OR

Write the cubes of the first five natural numbers and state which are odd. 3

18. Show that $(a \times b)^{-1} = a^{-1} \times b^{-1}$ for $a = 2,\ b = 5$. 3

OR

Write $\dfrac{4.2 \times 10^{5}}{2.1 \times 10^{2}}$ in scientific notation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A dance troupe forms a square arrangement with the same number of rows and columns. There are 1225 dancers. 5

(i) How many rows?

(ii) If 71 more join, can a new square be formed? Explain.

(iii) How many must be added to form the next square?

OR

(i) Estimate $\sqrt{200}$ between consecutive integers. 5

(ii) Find $\sqrt{200}$ to one decimal place using the nearest squares.

20. (i) Simplify $\dfrac{16^{2} \times 2^{3}}{4^{3}}$. 5

(ii) Express the result as a power of 2.

OR

Case study: A dealer's annual turnover is ₹ 4.8 crore. 5

(i) Write it in standard form.

(ii) If it grows by 10^{6} every year, find the turnover after 5 years.

(iii) Write the answer in standard form.

END OF QUESTION PAPER — Total: 40 Marks

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